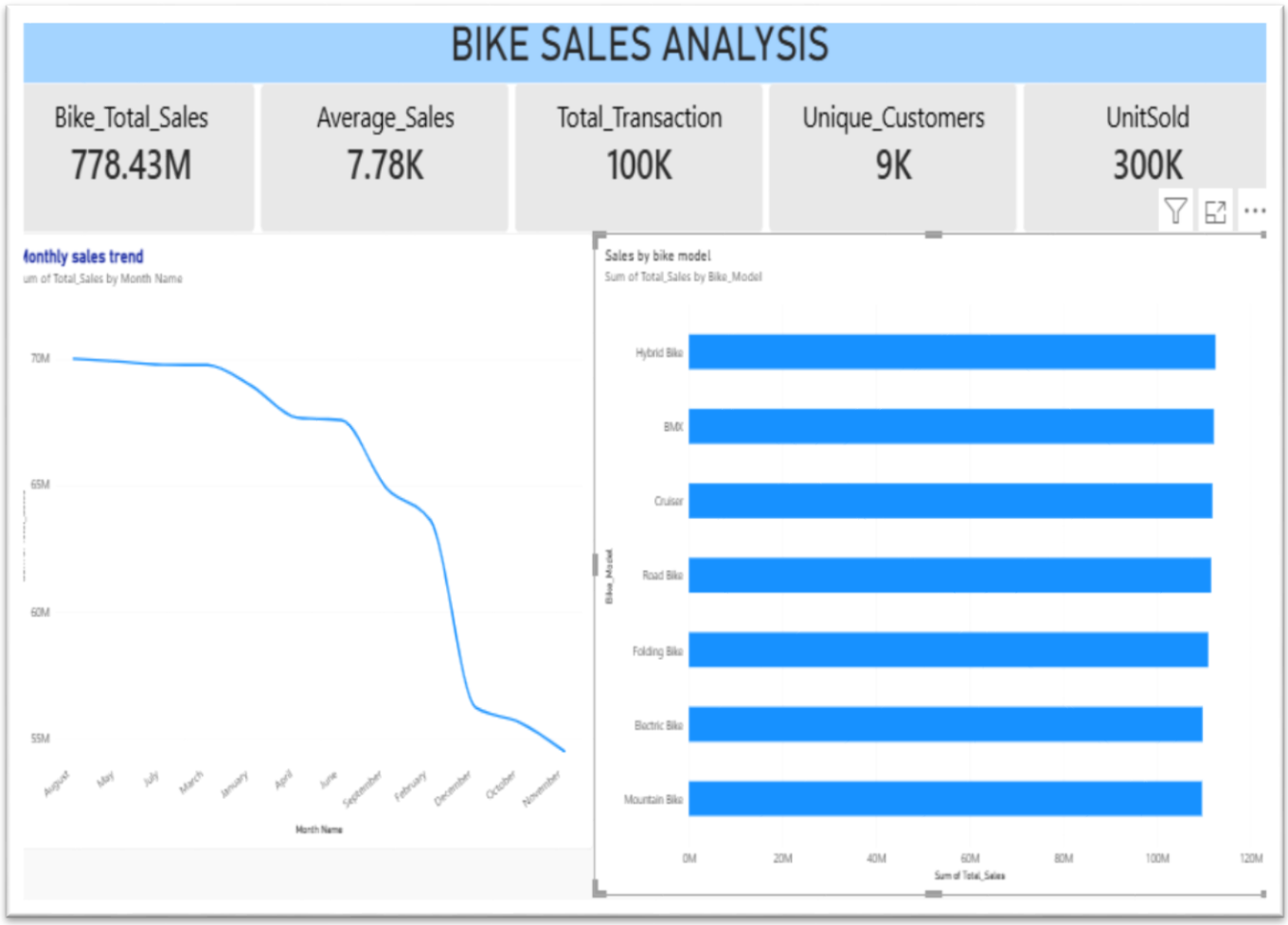
BIKE SALES ANALYSIS USING POWER BI

INTRODUCTION

This report presents a comprehensive evaluation of the **Bike Sales Analysis** dataset derived from Power BI reporting dashboards. The primary objective is to evaluate key operational metrics—including overall sales volume, customer transaction patterns, demographic breakdown, product model performance, regional distribution, and sales channels—to identify core growth drivers and operational risks.

By analyzing high-level KPIs alongside detailed breakdown visuals, this report provides strategic clarity on product positioning, seasonality, customer retention, and targeted marketing opportunities.



Executive Summary & Key Metrics

- **Total Sales: 778.43M**
- **Average Transaction Sales: 7.78K**
- **Total Transactions: 100K**
- **Unique Customers: 9K**
- **Units Sold: 300K**
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Key Dashboard Insights



1. Sales Performance by Bike Model & Pricing

- **Top Performing Models:** Hybrid Bike, BMX, Cruiser, Road Bike, and Folding Bike generate the highest total revenue (~110M+ each).
- **Price vs. Total Sales Breakdown:** Most bike models fall into a high-price segment ($6.0M–$6.3M total sum of price), driving the bulk of sales revenue. Electric Bikes show a significantly lower price-to-sales ratio ($5.8M sum of price), making them a niche or underperforming product category in overall revenue contribution.

2. Temporal Sales Trend

- **Declining Trend Line:** Sales revenue shows a steady decline throughout the year, dropping from a peak near **70M** (around August/May) down to **53M–55M** in late autumn (October/November).
- **Action Required:** Investigate seasonal demand changes, marketing strategy gaps, or inventory constraints occurring towards the end of the calendar year.

3. Geographical & Demographics Distribution

- **Store Locations:** Revenue is remarkably balanced across major retail locations:
  - **Top Cities:** New York leads slightly (~114M total sales), closely followed by Phoenix, Philadelphia, Houston, Chicago, Los Angeles, and San Antonio (~110M each).
- **Customer Age Groups:**
  - The **56–65**, **36–45**, **26–35**, and **46–55** age brackets represent the highest volume of total transactions (~19K transactions each).
  - Younger demographic (**18–25**, ~15K transactions) and seniors (**66+**, ~9.5K transactions) show lower overall engagement.

4. Payment Method Preferences

- The total visible share for top payment methods shows strong utilization of digital and card payments:
  - **PayPal / Debit Card / Apple Pay / Credit Card / Cash / Google Pay** are evenly distributed across transactions, with non-cash payment channels dominating total volume..

Recommended Action Items & Next Steps

1. **Seasonal Promotions:** Introduce targeted Q3/Q4 promotions and end-of-year holiday bundles to counter the late-year revenue decline.
2. **Growth Opportunities in Demographics:** Expand targeted marketing towards the 18–25 and 66+ age groups to tap into underperforming customer segments.
3. **Product Mix Optimization:** Re-evaluate pricing strategies and promotional visibility for lower-performing categories like Electric Bikes.

Conclusion

The Power BI dashboard highlights a strong overall sales foundation of **$778.43M** across **300K units sold**, backed by consistent distribution across key regional markets (led by New York) and core age demographics between 26 and 65. However, the data reveals critical vulnerabilities—

specifically a sharp late-year decline in monthly sales revenue and underperformance within the Electric Bike segment and fringe age groups (18–25 and 66+).

By executing the recommended seasonal campaigns, refining product pricing for lower-tier categories, and launching tailored marketing initiatives to underrepresented segments, the business can stabilize end-of-year sales and drive sustained profitability across all retail channels.